Warnaar's positivity conjecture at $d = 4$ (modulus 7): a complete proof

Seed 6, Layer 3, Round 2

Loop experiment, 2026-07-04

Setup

Fix $d = 4$, so $\ell = \gcd(d, 3) = 1$ and the modulus is $d + 3 = 7$. For a profile $c = (c_0, c_1, c_2)$ with $c_0 + c_1 + c_2 = 4$, let $F_c(z, q) = \sum_m F_{c,m}(q) z^m$ be the cylindric-partition generating function graded by largest part, and set

$$G_c(z, q) = (zq; q)_\infty F_c(z, q) = \sum_{n \geq 0} A_{n,c}(q)\, z^n, \qquad Q_{n,c}(q) = (q; q)_n A_{n,c}(q),$$

$$H_{c,m}(q) = (q; q)_m F_{c,m}(q).$$

Warnaar's Conjecture 2.7 at $d = 4$ asserts $Q_{n,c} \geq 0$ coefficientwise. We prove it for all 15 profiles, together with the Bounded Fermionic Form (BFF) statement and the monotonicity $H_{c,m} \geq H_{c,m-1}$.

Input from the literature. Corteel–Welsh (companion note to [CW19]; the note is fully proved by a uniqueness induction on a manifestly positive functional system) establish, in their profile labels, with $T(n, j) := q^{n^2+j^2-nj}$:

$$A_n^{(2,1,1)} = \sum_{j=0}^{2n} \frac{T(n,j)}{(q;q)_n} \begin{bmatrix} 2n \\ j \end{bmatrix}_q, \qquad A_n^{(4,0,0)} = \sum_j \frac{T(n,j)\, q^{n+j}}{(q;q)_n} \begin{bmatrix} 2n \\ j \end{bmatrix}_q, \qquad A_n^{(3,1,0)} = \sum_j \frac{T(n,j)\, q^j}{(q;q)_n} \begin{bmatrix} 2n \\ j \end{bmatrix}_q,$$

$$A_n^{(3,0,1)} = \sum_j T(n,j) \Big(\frac{q^n}{(q;q)_n} \begin{bmatrix} 2n \\ j \end{bmatrix}_q + \frac{q^{2j}}{(q;q)_{n-1}} \begin{bmatrix} 2n-2 \\ j \end{bmatrix}_q \Big),$$

$$A_n^{(2,2,0)} = \sum_j T(n,j) \Big(\frac{q^n}{(q;q)_n} \begin{bmatrix} 2n \\ j \end{bmatrix}_q + \frac{q^j(1+q^{n+j})}{(q;q)_{n-1}} \begin{bmatrix} 2n-2 \\ j \end{bmatrix}_q \Big),$$

together with the bounded (Euler-expanded) forms $F_{c,m} = \sum_{n \leq m} \widetilde{A}_{n,c}/(q;q)_{m-n}$, where $\widetilde{A}_{n,c}$ denotes the inner sums above (i.e. $A_{n,c}$ with the prefactors $1/(q;q)_n$, $1/(q;q)_{n-1}$ retained inside).

Orbit dictionary (corrected; computationally pinned, exact $\mathbb{Z}[q]$, $n \leq 28$). In the target-first convention of the problem statement (conjecture.tex) the C_3-orbits match the CW labels as

$$\begin{aligned} &\mathrm{CW}(2,1,1) = \{(1,1,2), (1,2,1), (2,1,1)\}, \quad \mathrm{CW}(4,0,0) = \{(0,0,4), (0,4,0), (4,0,0)\}, \\ &\mathrm{CW}(3,1,0) = \{(0,3,1), (1,0,3), (3,1,0)\}, \quad \mathrm{CW}(3,0,1) = \{(0,1,3), (1,3,0), (3,0,1)\}, \\ &\mathrm{CW}(2,2,0) = \{(0,2,2), (2,0,2), (2,2,0)\}. \end{aligned}$$

The last two are the "wall" orbits (no single positive double-sum form).

Convention note (erratum). An earlier version of this dictionary swapped the two chirality-sensitive rows CW$(3,1,0)$ and CW$(3,0,1)$: it was stated in the source-first (reversed) kernel convention used internally by the project's chain model, rather than the target-first kernel of conjecture.tex. In the corrected dictionary each CW label c maps to the orbit containing c itself (the identity map). Validation: independent adversarial recomputation (`seed8_R2L3_swapcheck.sage`, exact $\mathbb{Z}[q]$, $n \le 28$), reconfirmed by a separate verification agent; with the corrected dictionary all five orbits pass match/positivity/$Q_n(1) = 4^n$ checks. No mathematics in this note changes: the Q_n formulas, Absorption Lemmas A and B, and the main theorem are label-independent.

The Inversion Lemma

Lemma 1 (Inversion). *If $F_{c,m} = \sum_{n \le m} \widetilde{A}_n/(q;q)_{m-n}$ for polynomials $\widetilde{A}_n$ independent of m, then $Q_{n,c} = (q;q)_n \widetilde{A}_n$ and*

$$H_{c,m} = \sum_{n=0}^{m} \begin{bmatrix} m \\ n \end{bmatrix}_q Q_{n,c}.$$

Proof. Letting $m \to \infty$ coefficientwise and using Euler's theorem $\sum_k z^k/(q;q)_k = 1/(zq;q)_\infty$ (with $z \mapsto zq$ normalisation as in the grading), $(zq;q)_\infty F_c(z,q) = \sum_n \widetilde{A}_n z^n$, so $A_{n,c} = \widetilde{A}_n$ and $Q_{n,c} = (q;q)_n \widetilde{A}_n$. Then $H_{c,m} = (q;q)_m F_{c,m} = \sum_{n \le m} \frac{(q;q)_m}{(q;q)_{m-n}(q;q)_n}(q;q)_n \widetilde{A}_n = \sum_{n \le m} \begin{bmatrix} m \\ n \end{bmatrix}_q Q_{n,c}$. $\square$

Remark 2. *This is the inverse of the (verified) Q-transform $Q_n = \sum_m (-1)^{n-m} q^{\binom{n-m}{2}} \begin{bmatrix} n \\ m \end{bmatrix}_q H_{c,m}$ by Gauss q-binomial inversion, but the proof above is direct and needs no inversion theorem.*

Explicit Q_n and the wall absorption lemmas

By Lemma 1 applied to the CW forms (note $(q;q)_n/(q;q)_{n-1} = 1 - q^n$):

$$Q_n^{(2,1,1)} = \sum_j T(n,j) \begin{bmatrix} 2n \\ j \end{bmatrix}_q, \qquad Q_n^{(4,0,0)} = \sum_j T(n,j) q^{n+j} \begin{bmatrix} 2n \\ j \end{bmatrix}_q, \qquad Q_n^{(3,1,0)} = \sum_j T(n,j) q^j \begin{bmatrix} 2n \\ j \end{bmatrix}_q,$$
$$Q_n^{(3,0,1)} = X_n + (1-q^n) X'_n, \qquad Q_n^{(2,2,0)} = X_n + (1-q^n) Y'_n,$$

where

$$X_n = \sum_j T(n,j) q^n \begin{bmatrix} 2n \\ j \end{bmatrix}_q, \quad X'_n = \sum_j T(n,j) q^{2j} \begin{bmatrix} 2n-2 \\ j \end{bmatrix}_q, \quad Y'_n = \sum_j T(n,j) q^j (1+q^{n+j}) \begin{bmatrix} 2n-2 \\ j \end{bmatrix}_q.$$

The first three are manifestly nonnegative, with $Q_n(1) = \sum_j \binom{2n}{j} = 4^n$ as required. The walls need:

Lemma 3 (Absorption A). $X_n - q^n X'_n = \sum_j T(n,j) q^n \left((q^{j-1} + q^j) \begin{bmatrix} 2n-2 \\ j-1 \end{bmatrix}_q + \begin{bmatrix} 2n-2 \\ j-2 \end{bmatrix}_q \right) \ge 0.$

Proof. Two applications of the q-Pascal rule $\begin{bmatrix} N \\ j \end{bmatrix}_q = \begin{bmatrix} N-1 \\ j-1 \end{bmatrix}_q + q^j \begin{bmatrix} N-1 \\ j \end{bmatrix}_q$ give

$$\begin{bmatrix} 2n \\ j \end{bmatrix}_q = q^{2j} \begin{bmatrix} 2n-2 \\ j \end{bmatrix}_q + (q^{j-1} + q^j) \begin{bmatrix} 2n-2 \\ j-1 \end{bmatrix}_q + \begin{bmatrix} 2n-2 \\ j-2 \end{bmatrix}_q.$$

The first term contributes exactly $q^n\cdot$(the j-th summand of X_n'); the rest is nonnegative. $\square$

Lemma 4 (Absorption B)**.** $X_n - q^n Y_n' = \sum_{j=0}^{2n-1} q^{\,n^2+j^2-nj+2j+1} \begin{bmatrix} 2n-1 \\ j \end{bmatrix}_q \geq 0.$

Proof. Divide by the global factor q^{n^2+n} and set $D_n = \sum_j q^{j^2-nj} \begin{bmatrix} 2n \\ j \end{bmatrix}_q - \sum_j q^{j^2-nj}(q^j + q^{n+2j}) \begin{bmatrix} 2n-2 \\ j \end{bmatrix}_q$.

Step 1. The two q-Pascal rules $\begin{bmatrix} 2n \\ j \end{bmatrix}_q = \begin{bmatrix} 2n-1 \\ j-1 \end{bmatrix}_q + q^j \begin{bmatrix} 2n-1 \\ j \end{bmatrix}_q$ and $\begin{bmatrix} 2n-1 \\ j \end{bmatrix}_q = \begin{bmatrix} 2n-2 \\ j \end{bmatrix}_q + q^{2n-1-j} \begin{bmatrix} 2n-2 \\ j-1 \end{bmatrix}_q$ combine to

$$\begin{bmatrix} 2n \\ j \end{bmatrix}_q - q^j \begin{bmatrix} 2n-2 \\ j \end{bmatrix}_q = \begin{bmatrix} 2n-1 \\ j-1 \end{bmatrix}_q + q^{2n-1} \begin{bmatrix} 2n-2 \\ j-1 \end{bmatrix}_q, \tag{$*$}$$

valid for all $0 \leq j \leq 2n$ with the convention $\begin{bmatrix} N \\ k \end{bmatrix}_q = 0$ for $k < 0$ or $k > N$.

Step 2. Substituting ($*$),

$$D_n = \sum_j q^{j^2-nj} \begin{bmatrix} 2n-1 \\ j-1 \end{bmatrix}_q + \sum_j q^{j^2-nj+2n-1} \begin{bmatrix} 2n-2 \\ j-1 \end{bmatrix}_q - \sum_{j=0}^{2n-2} q^{j^2-nj+n+2j} \begin{bmatrix} 2n-2 \\ j \end{bmatrix}_q.$$

In the last sum put $j \mapsto j-1$: the exponent becomes $(j-1)^2 - n(j-1) + n + 2(j-1) = j^2 - nj + 2n - 1$, so it cancels the middle sum *exactly*. Hence $D_n = \sum_{j\geq 1} q^{j^2-nj} \begin{bmatrix} 2n-1 \\ j-1 \end{bmatrix}_q$; re-indexing $j \mapsto j+1$ and restoring q^{n^2+n} gives the claim. $\square$

Main theorem

Theorem 5 ($d = 4$ complete)**.** *For every profile c with $c_0 + c_1 + c_2 = 4$:*

(i) $Q_{n,c}(q) \geq 0$ coefficientwise for all n, with the manifestly positive formulas above (walls via Lemmas 3, 4); $Q_{n,c}(1) = 4^n$.

(ii) (BFF at $d = 4$) $H_{c,m} = \sum_{n\leq m} \begin{bmatrix} m \\ n \end{bmatrix}_q Q_{n,c}$ with $Q_{n,c} \geq 0$, i.e. the bounded fermionic-form coefficients are $a_n = Q_{n,c}$ for all *orbits, including the walls.*

(iii) (Monotonicity) $H_{c,m} - H_{c,m-1} = \sum_{n\geq 1} q^{m-n} \begin{bmatrix} m-1 \\ n-1 \end{bmatrix}_q Q_{n,c} \geq 0$, and hence $h_m = (H_m - H_{m-1}) + q^m H_{m-1} \geq 0$.

Proof. (i): the CW forms plus Lemmas 3 and 4. (ii): Lemma 1 applied to the CW bounded forms. (iii): the q-Pascal rule $\begin{bmatrix} m \\ n \end{bmatrix}_q - \begin{bmatrix} m-1 \\ n \end{bmatrix}_q = q^{m-n} \begin{bmatrix} m-1 \\ n-1 \end{bmatrix}_q$ applied termwise to (ii). $\square$

Remark 6 (Verification). *All ingredients were checked in exact* $\mathbb{Z}[q]$ *arithmetic (`scripts/seed6_R2L3_verify.py`): the* H*-recursion tower (*$m \le 8$*, all* 15 *profiles),* $Q_n \ge 0$ *and* $Q_n(1) = 4^n$ *(*$n \le 8$*), the orbit dictionary (*$n \le 7$*), the Inversion Lemma (*$m \le 8$*), the monotonicity identity, and both absorption identities (*$n \le 13$*). A brute-force enumeration check also revealed that the chain-model profile convention equals the* reversal *of the* H*-recursion convention:* $F_c^{\text{chain}} = F_{\text{rev}(c)}$, $\text{rev}(c_0, c_1, c_2) = (c_2, c_1, c_0)$.

Remark 7 (Beyond $d = 4$). *Lemma 1 is generic: whenever a CW-type bounded form (Euler expansion in* $1/(q;q)_{m-n}$*) exists, BFF and Monotonicity reduce to Q-positivity. The wall phenomenon is structural: walls satisfy* $Q_n = P_n + (1 - q^n)P'_n$ *with* $P_n, P'_n \ge 0$*, and positivity follows from an absorption inequality* $P_n \ge q^n P'_n$ *provable by q-Pascal manipulations (double-Pascal for one wall, shift-cancellation for the other).*